

Ques 1.) What are the different protocols you observe at the following layers of the protocol stack?

- A. Application layer
- B. Transport layer
- C. Network layer

Ans- During the packet capture in Wireshark while accessing <http://neverool.com> and <https://www.cornell.edu>, the following protocols were observed at different layers of the protocol stack.

Application layer -

The following application layer protocols were observed:-

- HTTP (HyperText Transfer Protocol) -

- It is used to access web pages and transfer data b/w the client and the server.

- DNS (Domain Name System) -

- Used to resolve the domain names into IP addresses before establishing the connection.

- TLS (Transport Layer Security) -

- Used to establish a secure encrypted connection while accessing <https://www.cornell.edu>

Transport layer -

The following transport layer protocols were observed,

- Tcp (Transmission Control Protocol) :-

- Used to provide reliable communication for HTTP and HTTPS traffic.

- UDP (User Datagram Protocol) :-

- Used mainly by DNS queries and responses because it is faster and suitable for small data transfers.

Network layer -

The following network layer protocol was observed:-

- IPv4 (Internet Protocol version-4) :-

- Used to route packets b/w source and destination IP addresses over the network.

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http

No.	Time	Source	Destination	Protocol	Length	Info
3472	35.392839300	10.85.23.124	34.223.124.45	HTTP	130	GET / HTTP/1.1
3505	35.951185700	34.223.124.45	10.85.23.124	HTTP	1514	[TCP Previous segment not captured] Continuation
3506	35.951189000	34.223.124.45	10.85.23.124	HTTP	1395	Continuation

Frame 3472: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits) on interface
Ethernet II, Src: Intel_93:e7:4d (90:65:84:93:e7:4d), Dst: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0)
Internet Protocol Version 4, Src: 10.85.23.124, Dst: 34.223.124.45
Transmission Control Protocol, Src Port: 57199, Dst Port: 80, Seq: 1, Ack: 1, Len: 76
Hypertext Transfer Protocol

0000 f8 e4 3b 35 2d a0 90 65 84 93 e7 4d 08 00 45 00 ...;5...e...M...E...
0010 00 74 43 23 40 00 80 06 00 00 0a 55 17 7c 22 df ...tC#@...U...|...
0020 7c 2d df 6f 00 50 03 ff 58 1f 68 a7 44 62 50 18 |-oP...X:hDbP...
0030 ff ff c1 43 00 00 47 45 54 20 2f 20 48 54 54 50 ...C...GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 /1.1..Ho st: neve
0050 72 73 73 6c 2e 63 6f 6d 0d 0a 55 73 65 72 2d 41 rssl.com ..User-A
0060 67 65 6e 74 3a 20 63 75 72 6c 2f 38 2e 32 31 2e gent: cu rl/8.21.
0070 30 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0 ..Accep t: /*...

Hypertext Transfer Protocol: Protocol

Packets: 8734 · Displayed: 3 (0.0%) · Dropped: 0 (0.0%) Profile: Default

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
2970	29.414528900	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14311 Win=2116 Len=0
2971	29.435304500	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14401 Win=2127 Len=0 SLE=14581 SRE=14671
2972	29.435305600	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14491 Win=2127 Len=0 SLE=14581 SRE=14671
2973	29.435307000	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14671 Win=2127 Len=0
2974	29.435307400	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14401 Win=2116 Len=0
2975	29.435307800	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14671 Win=2138 Len=0 SLE=14761 SRE=14851
2976	29.435308200	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14851 Win=2138 Len=0
2977	29.435308700	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14942 Win=2138 Len=0
2998	29.704043500	57.144.143.32	10.85.23.124	TLSv1.2	125	Application Data
2999	29.755911200	10.85.23.124	57.144.143.32	TCP	54	62037 → 443 [ACK] Seq=14942 Ack=244 Win=511 Len=0
3054	30.391171400	10.85.23.124	23.201.59.97	TCP	54	57878 → 80 [FIN, ACK] Seq=1 Ack=1 Win=251 Len=0
3062	30.424889400	23.201.59.97	10.85.23.124	TCP	64	80 → 57878 [FIN, ACK] Seq=1 Ack=2 Win=501 Len=0
3063	30.424964900	10.85.23.124	23.201.59.97	TCP	54	57878 → 80 [ACK] Seq=2 Ack=2 Win=251 Len=0
3148	31.648411100	10.85.23.124	23.208.64.169	TCP	55	60915 → 443 [ACK] Seq=1 Ack=1 Win=250 Len=1
3163	31.664335200	23.208.64.169	10.85.23.124	TCP	66	443 → 60915 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
3196	32.060108900	10.85.23.124	34.223.124.45	TCP	66	57199 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK PERM
3296	33.065042900	10.85.23.124	34.223.124.45	TCP	66	[TCP Retransmission] 57199 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK PERM
3366	34.103388100	10.85.23.124	23.197.216.204	TLSv1.2	106	Application Data
3371	34.121587800	23.197.216.204	10.85.23.124	TCP	60	443 → 64524 [ACK] Seq=1658 Ack=181 Win=501 Len=0
3372	34.125959900	23.197.216.204	10.85.23.124	TLSv1.2	228	Application Data
3373	34.125961200	23.197.216.204	10.85.23.124	TLSv1.2	172	Application Data
3374	34.126036600	10.85.23.124	23.197.216.204	TCP	54	64524 → 443 [ACK] Seq=181 Ack=1950 Win=253 Len=0
3443	35.030874800	54.192.151.112	10.85.23.124	TLSv1.2	78	Application Data
3444	35.030875300	54.192.151.112	10.85.23.124	TCP	60	443 → 60916 [FIN, ACK] Seq=25 Ack=1 Win=136 Len=0
3445	35.031379300	10.85.23.124	54.192.151.112	TCP	54	60916 → 443 [ACK] Seq=1 Ack=26 Win=255 Len=0
3446	35.076188400	10.85.23.124	34.223.124.45	TCP	66	[TCP Retransmission] 57199 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK PERM
3470	35.392297000	34.223.124.45	10.85.23.124	TCP	60	80 → 57199 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460
3471	35.392482500	10.85.23.124	34.223.124.45	TCP	54	57199 → 80 [ACK] Seq=1 Ack=1 Win=65535 Len=0

Frame 3163: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...

Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_93:e7:4d (90:65:84:93:...

Internet Protocol Version 4, Src: 23.208.64.169, Dst: 10.85.23.124

Transmission Control Protocol, Src Port: 443, Dst Port: 60915, Seq: 1, Ack: 2, Len: 0

0000 90 65 84 93 e7 4d f8 e4 3b 35 2d a0 08 00 45 00 e...M...;5...E

0010 00 34 41 eb 40 00 3a 06 84 8f 17 d0 40 a9 0a 55 4A @ : . . . @ . U

0020 17 7c 01 bb ed f3 61 f1 76 c9 52 be 4a 0a 80 10 | . . . a v R J . . .

0030 01 f5 5e bc 00 00 01 01 05 0a 52 be 4a 09 52 be R J R .

0040 4a 0a J .

Transmission Control Protocol: Protocol

Packets: 8734 · Displayed: 1146 (13.1%) · Dropped: 0 (0.0%) Profile: Default

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Wireshark interface showing a packet capture of DNS traffic. The packet list pane displays 27 packets, with packet 149 selected. The packet details pane shows the structure of the selected packet, including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response). The packet bytes pane shows the raw data in hexadecimal and ASCII.

Domain Name System: Protocol

Packets: 8734 · Displayed: 22 (0.3%) · Dropped: 0 (0.0%) Profile: Default

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tls

No.	Time	Source	Destination	Protocol	Length	Info
2908	29.179848600	10.85.23.124	57.144.143.32	TLSv1.2	178	Application Data
2909	29.179889400	10.85.23.124	57.144.143.32	TLSv1.2	158	Application Data
2910	29.179930300	10.85.23.124	57.144.143.32	TLSv1.2	160	Application Data
2911	29.179968000	10.85.23.124	57.144.143.32	TLSv1.2	159	Application Data
2912	29.180031800	10.85.23.124	57.144.143.32	TLSv1.2	196	Application Data
2949	29.378984800	10.85.23.124	57.144.143.32	TLSv1.2	212	Application Data
2950	29.390541200	10.85.23.124	57.144.143.32	TLSv1.2	123	Application Data
2951	29.393036300	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2952	29.393144900	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2953	29.393222100	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2957	29.393273200	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2958	29.393308000	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2959	29.393347700	10.85.23.124	57.144.143.32	TLSv1.2	144	Application Data
2960	29.393396800	10.85.23.124	57.144.143.32	TLSv1.2	145	Application Data
2998	29.704043500	57.144.143.32	10.85.23.124	TLSv1.2	125	Application Data
3366	34.103388100	10.85.23.124	23.197.216.204	TLSv1.2	106	Application Data
3372	34.125959900	23.197.216.204	10.85.23.124	TLSv1.2	228	Application Data
3373	34.125961200	23.197.216.204	10.85.23.124	TLSv1.2	172	Application Data
3443	35.030874800	54.192.151.112	10.85.23.124	TLSv1.2	78	Application Data
3776	38.793175200	10.85.23.124	23.197.216.204	TLSv1.2	106	Application Data
3777	38.812232600	23.197.216.204	10.85.23.124	TLSv1.2	229	Application Data
3778	38.812243400	23.197.216.204	10.85.23.124	TLSv1.2	765	Application Data
3974	41.274668200	172.64.148.235	10.85.23.124	TLSv1.2	78	Application Data
3975	41.275164700	10.85.23.124	172.64.148.235	TLSv1.2	82	Application Data
4089	42.831582300	10.85.23.124	23.197.216.204	TLSv1.2	106	Application Data
4091	42.848293300	23.197.216.204	10.85.23.124	TLSv1.2	307	[TCP Previous segment not captured], Application Data
4509	46.727868300	10.85.23.124	23.197.216.204	TLSv1.2	106	Application Data
4510	46.745160200	23.197.216.204	10.85.23.124	TLSv1.2	229	Application Data
4511	46.745160200	23.197.216.204	10.85.23.124	TLSv1.2	229	Application Data

Frame 2998: Packet, 125 bytes on wire (1000 bits), 125 bytes captured (1000 bits) on interface 0000 90 65 84 93 e7 4d f8 e4 3b 35 2d a0 08 00 45 00 e...M...;5...E...
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_93:e7:4d (90:65:84:93: 0010 00 6f 0b fb 40 00 57 06 2d 0d 39 90 8f 20 0a 55 o...@W...-9...U...
Internet Protocol Version 4, Src: 57.144.143.32, Dst: 10.85.23.124 0020 17 7c 01 bb f2 55 6d 58 92 39 3a 6f da a9 50 18 [...UmX...9:o...P...
Transmission Control Protocol, Src Port: 443, Dst Port: 62037, Seq: 173, Ack: 14942, Len: 71 0030 08 5a f7 c1 00 00 17 03 03 00 42 d8 19 36 3c f2 Z...]...-B...6<...
Transport Layer Security 0040 af 20 e1 12 5d 99 ba a4 d1 fe 09 b0 9f a4 25 ff ...]...%...
0050 b0 bc 90 4b 7a d8 7a d2 e9 cc a7 85 05 58 18 0c ...Kz z...X...
0060 49 70 1c f4 cf 0c f9 ad ea 7b 04 f7 ed 2c 10 11 Ip...{...X...
0070 f4 e0 8b 95 b4 04 23 43 c6 13 65 58 3b#C...eX;

Transport Layer Security: Protocol

Packets: 8734 · Displayed: 423 (4.8%) · Dropped: 0 (0.0%) Profile: Default

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ip

No.	Time	Source	Destination	Protocol	Length	Info
2969	29.398333400	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14242 Win=2116 Len=0
2970	29.414528900	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14311 Win=2116 Len=0
2971	29.435304500	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14401 Win=2127 Len=0 SLE=14581 SRE=14671
2972	29.435305600	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14491 Win=2127 Len=0 SLE=14581 SRE=14671
2973	29.435307000	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14671 Win=2127 Len=0
2974	29.435307400	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14401 Win=2116 Len=0
2975	29.435307800	57.144.143.32	10.85.23.124	TCP	66	443 → 62037 [ACK] Seq=173 Ack=14671 Win=2138 Len=0 SLE=14761 SRE=14851
2976	29.435308200	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14851 Win=2138 Len=0
2977	29.435308700	57.144.143.32	10.85.23.124	TCP	60	443 → 62037 [ACK] Seq=173 Ack=14942 Win=2138 Len=0
2981	29.505427300	10.85.18.1	255.255.255.255	UDP	864	40909 → 29810 Len=822
2983	29.505428800	10.85.22.203	255.255.255.255	UDP	868	49234 → 29810 Len=826
2985	29.601169600	10.85.17.213	255.255.255.255	UDP	867	42942 → 29810 Len=825
2988	29.601172700	10.85.20.217	224.0.0.251	MDNS	107	Standard query response 0x0000 PTR I0YyNUv8n14AAA._FC9F5ED42C8A._tcp.local
2994	29.704039500	10.85.17.79	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
2998	29.704043500	57.144.143.32	10.85.23.124	TLSv1.2	125	Application Data
2999	29.755911200	10.85.23.124	57.144.143.32	TCP	54	62037 → 443 [ACK] Seq=14942 Ack=244 Win=511 Len=0
3002	29.811335100	10.85.20.155	239.255.255.250	UDP/XML	666	62832 → 3702 Len=624
3009	30.016741500	0.0.0.0	255.255.255.255	DHCP	365	DHCP Request - Transaction ID 0x1f7b9a69
3011	30.016744100	10.85.16.122	255.255.255.255	UDP	866	57145 → 29810 Len=824
3014	30.016745700	10.85.17.169	255.255.255.255	UDP	867	59507 → 29810 Len=825
3017	30.016747400	10.85.18.18	255.255.255.255	UDP	865	39486 → 29810 Len=823
3021	30.016758100	10.85.17.79	10.85.23.255	NBNS	110	Registration NB DESKTOP-JCR4914<00>
3022	30.016758500	10.85.21.204	255.255.255.255	TPLINK...	100	UDP Cmd: {"system":{"get_sysinfo":{}},"emeter":{"get_realtime":{}}}
3023	30.023117000	10.85.17.134	224.0.0.252	LLMNR	75	Standard query 0x8440 ANY LAPTOP-6075BL44
3030	30.119385700	10.85.20.155	224.0.0.251	MDNS	75	Standard query 0x0000 PTR _ipp._tcp.local, "QM" question
3032	30.119386800	10.85.17.79	10.85.23.255	NBNS	110	Registration NB WORKGROUP<00>
3036	30.215699600	104.18.41.41	10.85.23.124	UDP	76	443 → 50455 Len=34
3037	30.234491100	10.85.23.124	104.18.41.41	UDP	89	50455 → 443 Len=47

Transport Layer Security

[Stream index: 2]

TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

Content Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 66

Encrypted Application Data: d819363cf2af20e1125d99baa4d1fe09b09fa425ffb0bc904b7ad87ad2

Payload is encrypted application data (tls.app_data), 66 bytes

Packets: 8734 · Displayed: 4200 (48.1%) · Dropped: 0 (0.0%) Profile: Default

30°C Partly cloudy

Search

ENG IN

8:00 PM 8/1/2026

Ques 2) What is the total amount of data being received for the following two cases?

A. When you access <http://neverssl.com/>

Ans. Total amount of data received while accessing <http://neverssl.com> = 2977 bytes

B. When you access <https://www.Cornell.edu>

Ans:- The total ~~amount~~ amount of data received with <https://www.Cornell.edu> = 95KB

- Byte $A \rightarrow B$ = 3KB (Client $\rightarrow$ Server, request)
- Byte $B \rightarrow A$ = 92KB (Server $\rightarrow$ Client, recieved data)

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tcp.stream eq 26

No.	Time	Source
3196	32.060108900	10.85.23.124
3296	33.065042900	10.85.23.124
3446	35.076188400	10.85.23.124
3470	35.392297000	34.223.124.45
3471	35.392482500	10.85.23.124
3472	35.392839300	10.85.23.124
3498	35.745821300	34.223.124.45
3505	35.951185700	34.223.124.45
3506	35.951189000	34.223.124.45
3507	35.951190800	34.223.124.45
3508	35.951230800	10.85.23.124
3509	35.951289900	10.85.23.124
3510	35.951346700	10.85.23.124
3511	35.952633800	10.85.23.124
3524	36.256438200	34.223.124.45
3525	36.256510000	10.85.23.124

Acknowledgment Number: 77
Acknowledgment number (range) 0101 = Header Length
Flags: 0x010 (ACK)
Window: 29200
[Calculated window size:
[Window size scaling factor
Checksum: 0x9c89 [unverified
[Checksum Status: Unverified
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Stream
[Server Contiguous Stream
TCP payload (1460 bytes)
Hypertext Transfer Protocol
File Data: 1460 bytes
Data (1460 bytes)

Specifies if this is an individual packet

Wireshark · Follow HTTP Stream (tcp.stream eq 26) · CN_lab02.pcapng

on a wifi network, and nothing happens. Type "http://neverssl.com" into your browser's url bar, and you'll be able to log on.</p>

<h2>How?</h2>
<p>neverssl.com will never use SSL (also known as TLS). No encryption, no strong authentication, no HSTS, no HTTP/2.0, just plain old unencrypted HTTP and forever stuck in the dark ages of internet security.</p>

<h2>Why?</h2>
<p>Normally, that's a bad idea. You should always use SSL and secure encryption when possible. In fact, it's such a bad idea that most websites are now using https by default.</p>

<p>And that's great, but it also means that if you're relying on poorly-behaved wifi networks, it can be hard to get online. Secure browsers and websites using https make it impossible for those wifi networks to send you to a login or payment page. Basically, those networks can't tap into your connection just like attackers can't. Modern browsers are so good that they can remember when a website supports encryption and even if you type in the website name, they'll use https.</p>

<p>And if the network never redirects you to this page, well as you can see, you're not missing much.</p>

Follow @neverssl

</noscript>
</div>
</div>
</body>
</html>

1 client pkt, 2 server pkts, 1 turn.

Entire conversation (2877 bytes)
34.223.124.45:80 → 10.85.23.124:57199 (2801 bytes)
10.85.23.124:57199 → 34.223.124.45:80 (76 bytes)

Show as ASCII No delta times Stream 26 Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

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Wireshark · Conversations · CN_lab02.pcapng

Conversation Settings

☐ Name resolution

☒ Absolute start time

☒ Display raw data

☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

Bluetooth

BPv7

DCCP

DNP 3.0

☒ Ethernet

FC

FDDI

IEEE 802.11

Filter list for specific type

Ethernet · 1

IPv4 · 1

IPv6

TCP · 1

UDP

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
10.85.23.124	50061	150.171.109.131	443	89	95 kB	51	89	100.00%	41	3 kB	48	

Source Port: 50061

Destination Port: 443

[Stream index: 51]

[Stream Packet Number: 1]

[Conversation completeness: Complete, WITH DATA (31)]

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Packets: 8734 · Displayed: 89 (1.0%) · Dropped: 0 (0.0%)

Profile: Default

30°C

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Search

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15%

8:59 PM

8/1/2026

Ques 2) Answer the following when you access the site <http://never881.com/>.

A. How many HTTP GET requests do you observe?
List down the GET requests.

Ans - Total HTTP GET Requests = 1

GET Request :- GET /HTTP/1.1

B. What version of HTTP the server is running?

Ans - HTTP version = HTTP/1.1

C. For each of the HTTP GET requests, find out :-

i) The total number of TCP segments being received.

Ans - 3 TCP segments were received from the server.

ii) Total amount of data received in the corresponding HTTP response. -

Ans - Content-length = 3961 bytes

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http

No.	Time	Source	Destination	Protocol	Length	Info
3472	35.392839300	10.85.23.124	34.223.124.45	HTTP	130	GET / HTTP/1.1
3505	35.951185700	34.223.124.45	10.85.23.124	HTTP	1514	[TCP Previous segment not captured] Continuation
3506	35.951189000	34.223.124.45	10.85.23.124	HTTP	1395	Continuation

Frame 3472: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits) on interface
Ethernet II, Src: Intel_93:e7:4d (90:65:84:93:e7:4d), Dst: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0)
Internet Protocol Version 4, Src: 10.85.23.124, Dst: 34.223.124.45
Transmission Control Protocol, Src Port: 57199, Dst Port: 80, Seq: 1, Ack: 1, Len: 76
Hypertext Transfer Protocol

0000 f8 e4 3b 35 2d a0 90 65 84 93 e7 4d 08 00 45 00 ...;5...e...M...E...
0010 00 74 43 23 40 00 80 06 00 00 0a 55 17 7c 22 df ...tC#@...U...|...
0020 7c 2d df 6f 00 50 03 ff 58 1f 68 a7 44 62 50 18 |-oP...X:hDbP...
0030 ff ff c1 43 00 00 47 45 54 20 2f 20 48 54 54 50 ...C...GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 /1.1...Ho st: neve
0050 72 73 73 6c 2e 63 6f 6d 0d 0a 55 73 65 72 2d 41 rssl.com...User-A
0060 67 65 6e 74 3a 20 63 75 72 6c 2f 38 2e 32 31 2e gent: cu rl/8.21.
0070 30 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0...Accep t: /*...

Hypertext Transfer Protocol: Protocol

Packets: 8734 · Displayed: 3 (0.0%) · Dropped: 0 (0.0%) Profile: Default

30°C Partly cloudy

7:57 PM 8/1/2026

CN_lab02.pcapng

File Edit View Go Capture An

tcp.stream eq 26

No.	Time	Source
3196	32.060108900	10.85...
3296	33.065042900	10.85...
3446	35.076188400	10.85...
3470	35.392297000	34.223...
3471	35.392482500	10.85...
3472	35.392839300	10.85...
3498	35.745821300	34.223...
3505	35.951105700	34.223...
3506	35.951189000	34.223...
3507	35.951190800	34.223...
3508	35.951230800	10.85...
3509	35.951289900	10.85...
3510	35.951346700	10.85...
3511	35.952633800	10.85...
3524	36.256438200	34.223...
3525	36.256510000	10.85...

Frame 3472: Packet, 130 byte

- Ethernet II, Src: Intel 93:e6:9d:80:5a:13, Dst: 08:00:27:00:00:00
- Internet Protocol Version 4, Src: 10.85.1.1, Dst: 10.85.1.2
- Transmission Control Protocol, Src Port: 4444, Dst Port: 80
- Hypertext Transfer Protocol
 - GET / HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /
 - Request Version: HTTP/1.1
 - Host: neverssl.com\r\n
 - User-Agent: curl/8.21.0\r\n
 - Accept: */*\r\n
 - \r\n
 - [Full request URI: http://neverssl.com/]

Wireshark · Follow TCP Stream (tcp.stream eq 26) · CN_lab02.pcapng

GET / HTTP/1.1
Host: neverssl.com
User-Agent: curl/8.21.0
Accept: */*

HTTP/1.1 200 OK
Date: Sat, 01 Aug 2026 14:24:35 GMT
Server: Apache/2.4.66 ()
Upgrade: h2,h2c
Connection: Upgrade
Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT
ETag: "f79-5e28b29d38e93"
Accept-Ranges: bytes
Content-Length: 3961
Vary: Accept-Encoding
Content-Type: text/html; charset=UTF-8

<html>
 <head>
 <title>NeverSSL - Connecting ... </title>
 <style>
 body {
 font-family: Montserrat, helvetica, arial, sans-serif;
 font-size: 16px;
 color: #444444;
 margin: 0;
 }
 h2 {
 font-weight: 700;
 font-size: 1.6em;
 margin-top: 30px;
 }
 p {
 line-height: 1.6em;
 }
 .container {

1 client pkt, 3 server pkts, 1 turn.

Entire conversation (1536 bytes) Show as ASCII No delta times Stream 26

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Details

010101
011010
011100

CN_lab02

Wireshark capture file
2.39 MB
C:\Users\DELL\OneDrive\Doc...
8/1/2026 7:56 PM
Sync pending

Properties

Q. Identify TCP three way handshake used to establish the connection with the `http://never881.com`. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Ans.

<u>Packet</u>	<u>Flags</u>	<u>Seq. no.</u>	
Packet 3196	SYN	0	
Packet 3470	SYN, ACK	0	
Client ACK	ACK	1	